Reconocimiento de Patrones

Versión 2026

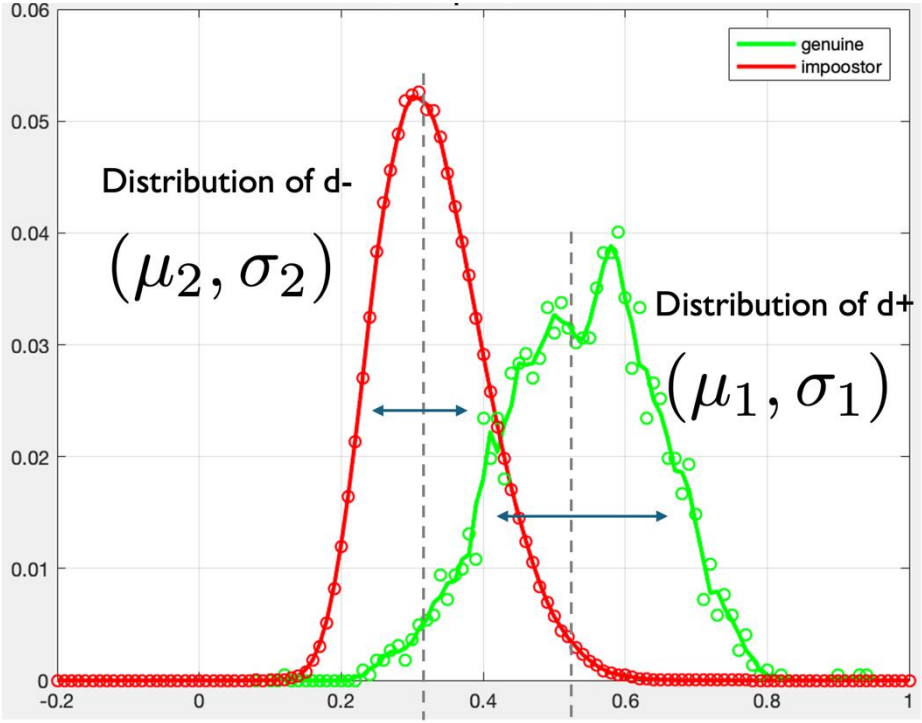


Performance

(CAPÍTULO 5 – PR.05.B)

Domingo Mery

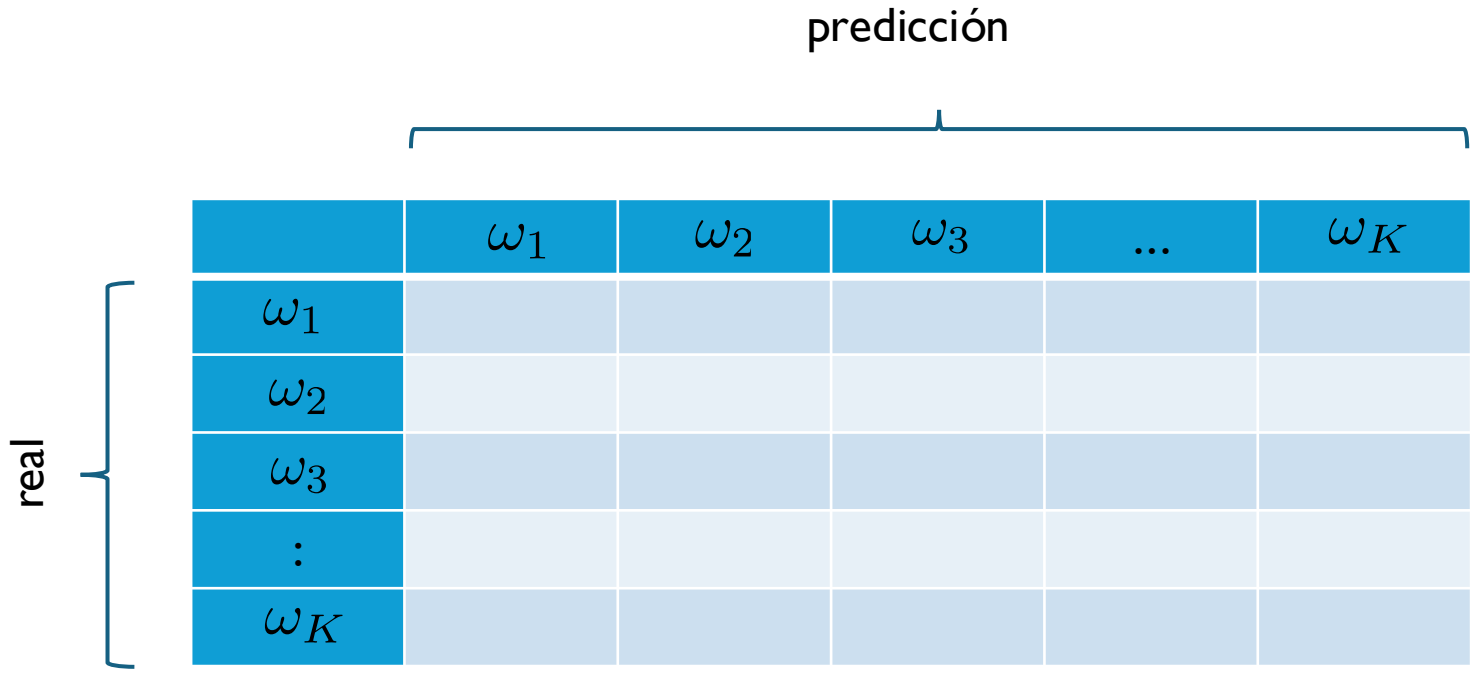
Ciencia de la Computación
Universidad Católica – Chile
http://domingomery.ing.uc.cl



Matriz de Confusión

The confusion matrix, $\mathbf{T}$, is a $K \times K$ matrix, where K is the number of classes of our data. The element $T(i, j)$ of the confusion matrix is defined as the number of samples that belong to class ω_i and were classified as ω_j. A perfect classification means that $T(i, i)$ is N_i and $T(i, j) = 0$ for $i \neq j$, where N_i is the number of samples of class ω_i.

Matriz de Confusión



Matriz de Confusión

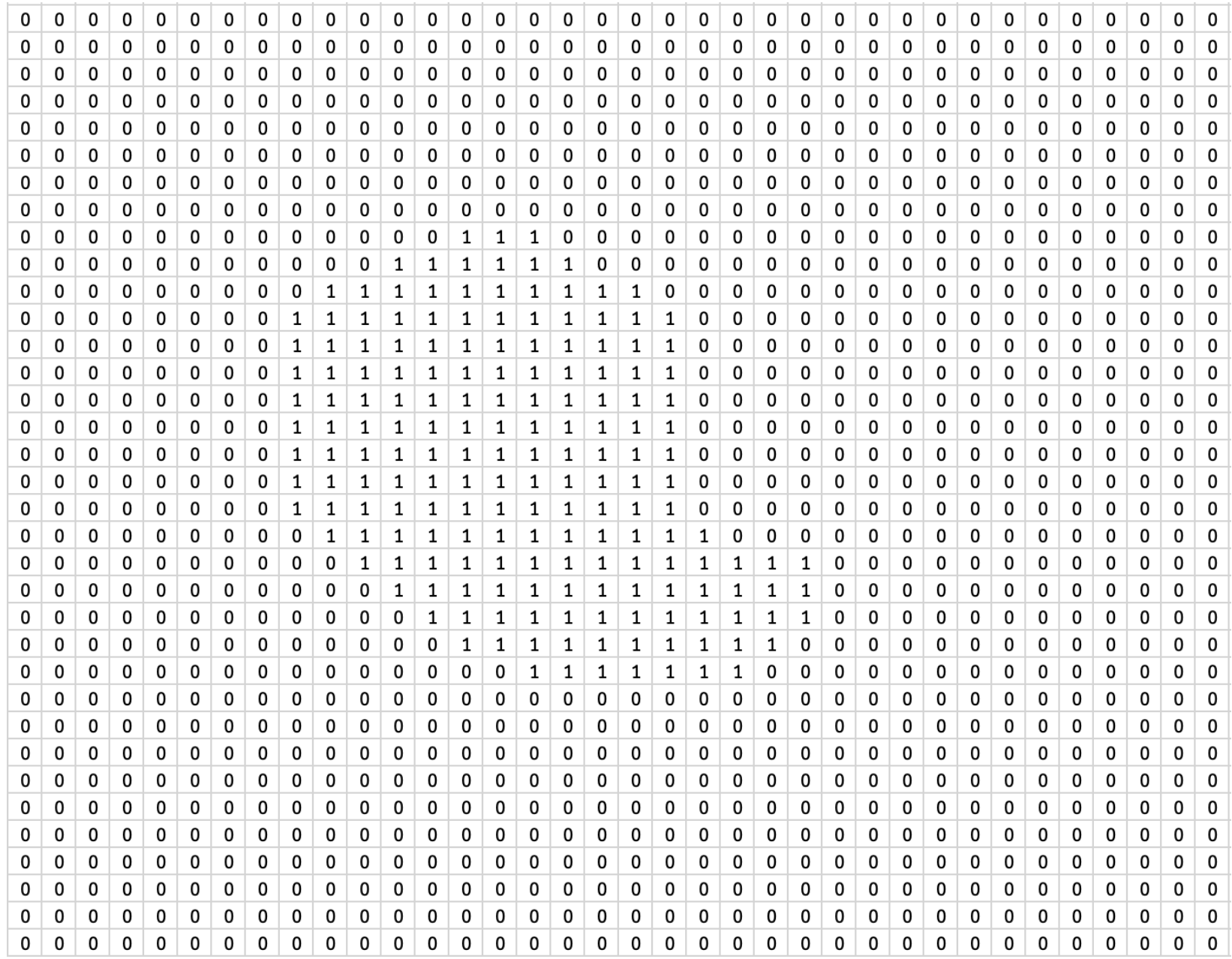
predicción

real

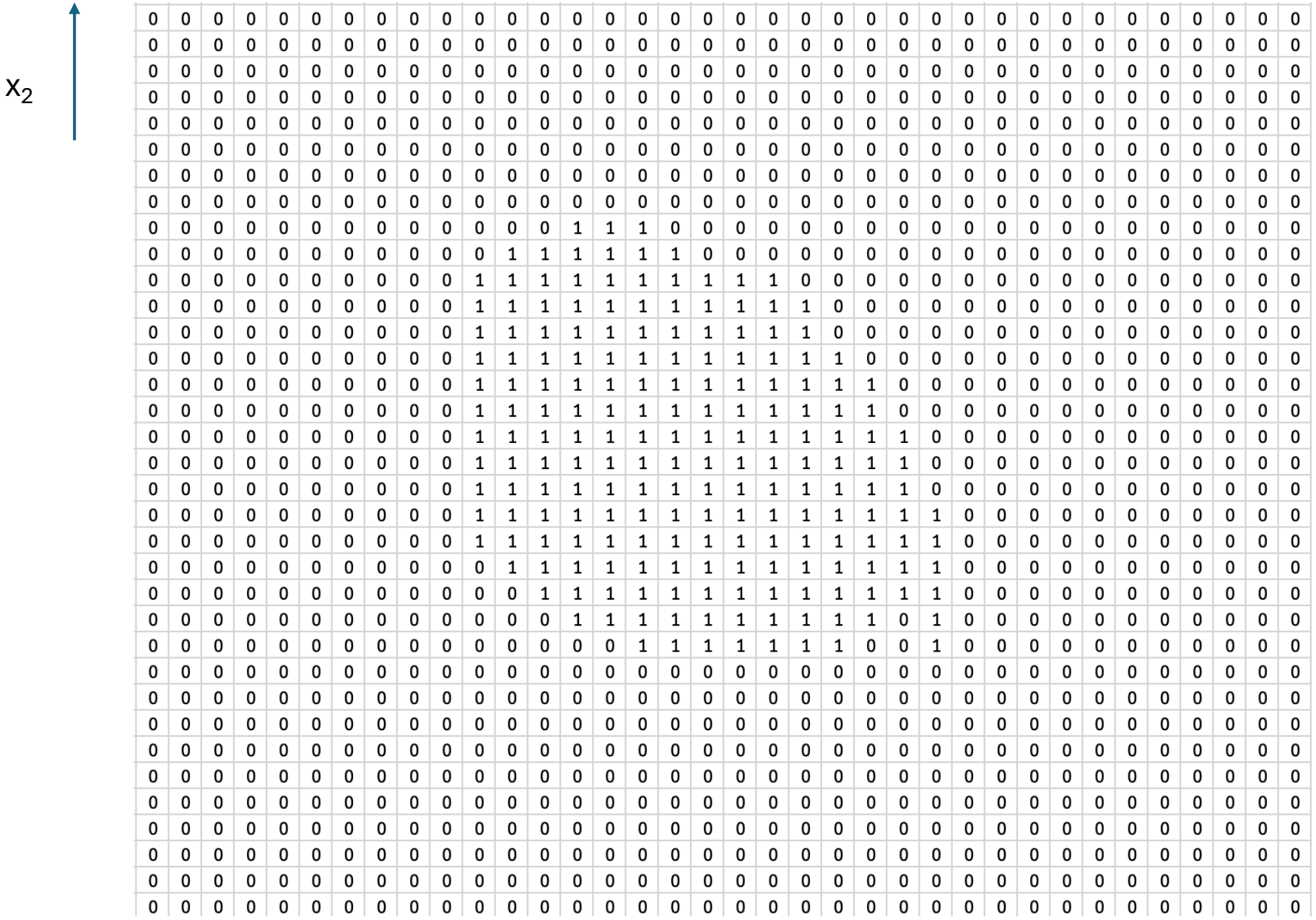
	ω_1	ω_2	ω_3	...	ω_K
ω_1	1230	90	78		9
ω_2	66	890	120		5
ω_3	59	95	1527		90
:					
ω_K	40	67	129		912

Ejemplo: hay 78 muestras de la clase 1 que han sido clasificadas como clase 3.

Detección
Problema de 2 clases

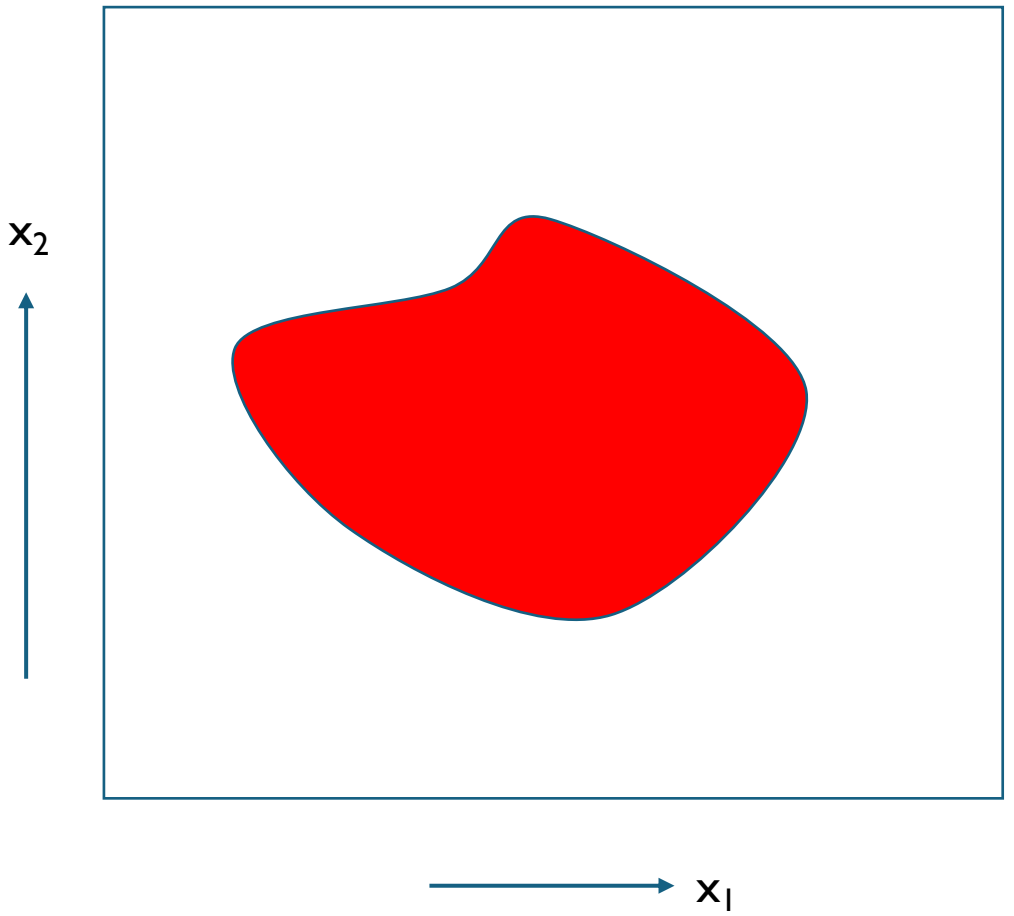


Ground Truth



Detección

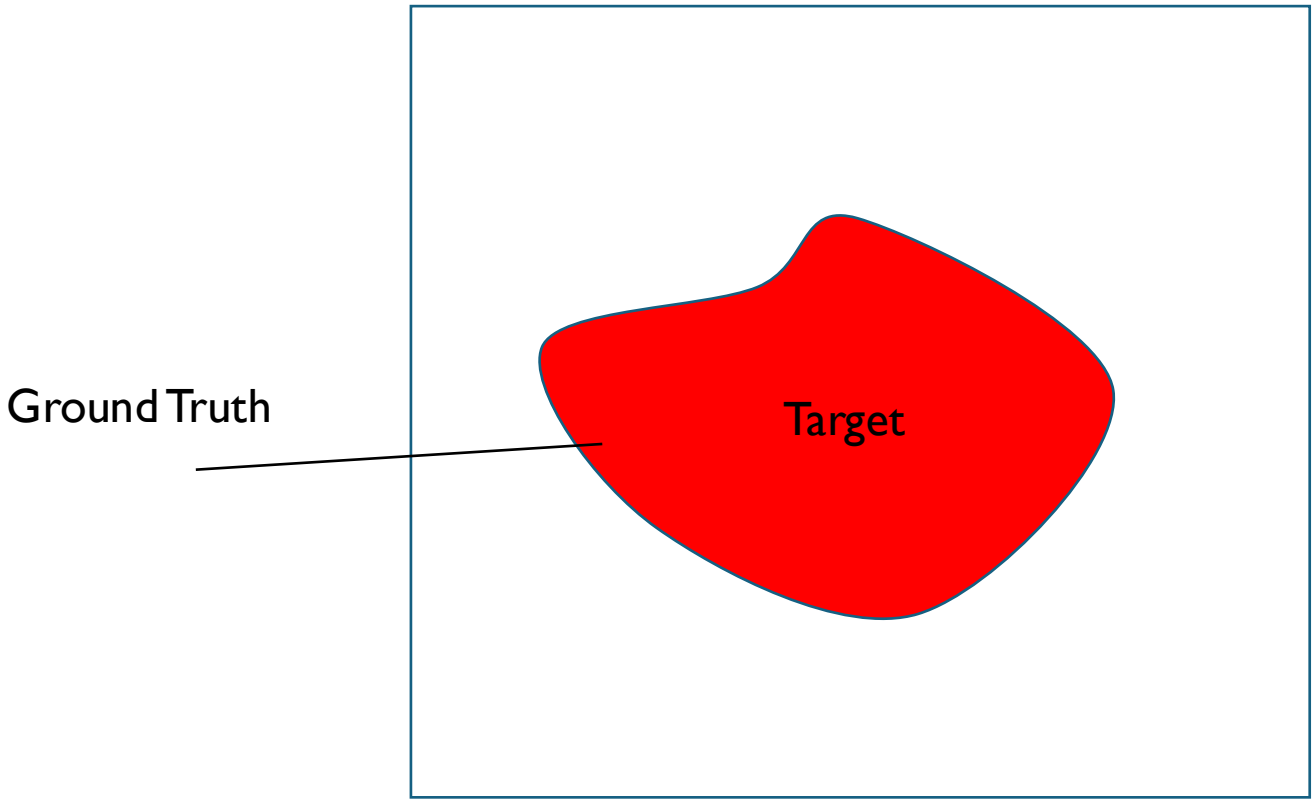
Ejemplo



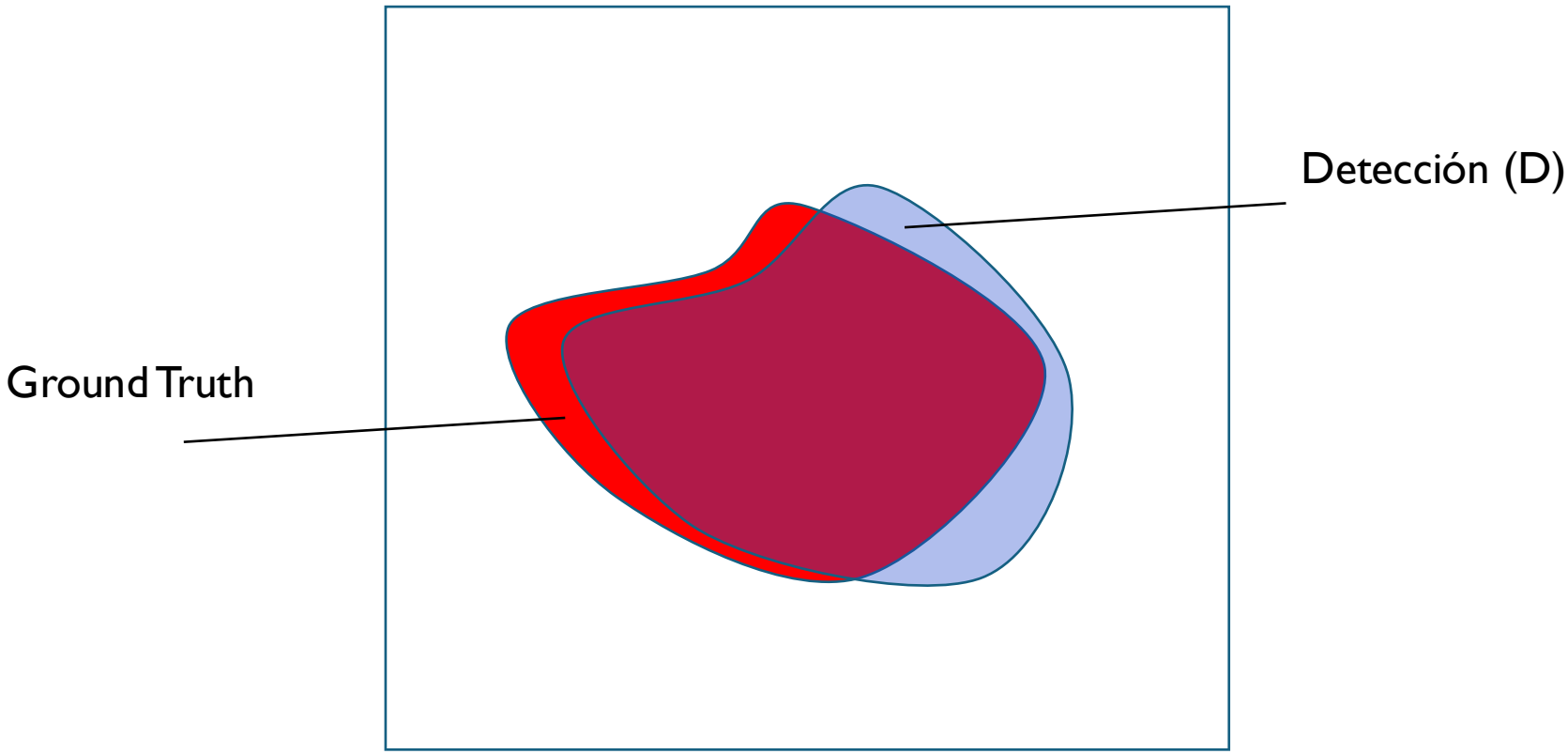
class 1
(clase positive – Ground Truth)

Ejemplo

Píxeles rojos: Muestras positivas (P)
Píxeles blancos: Muestras negativas (N)

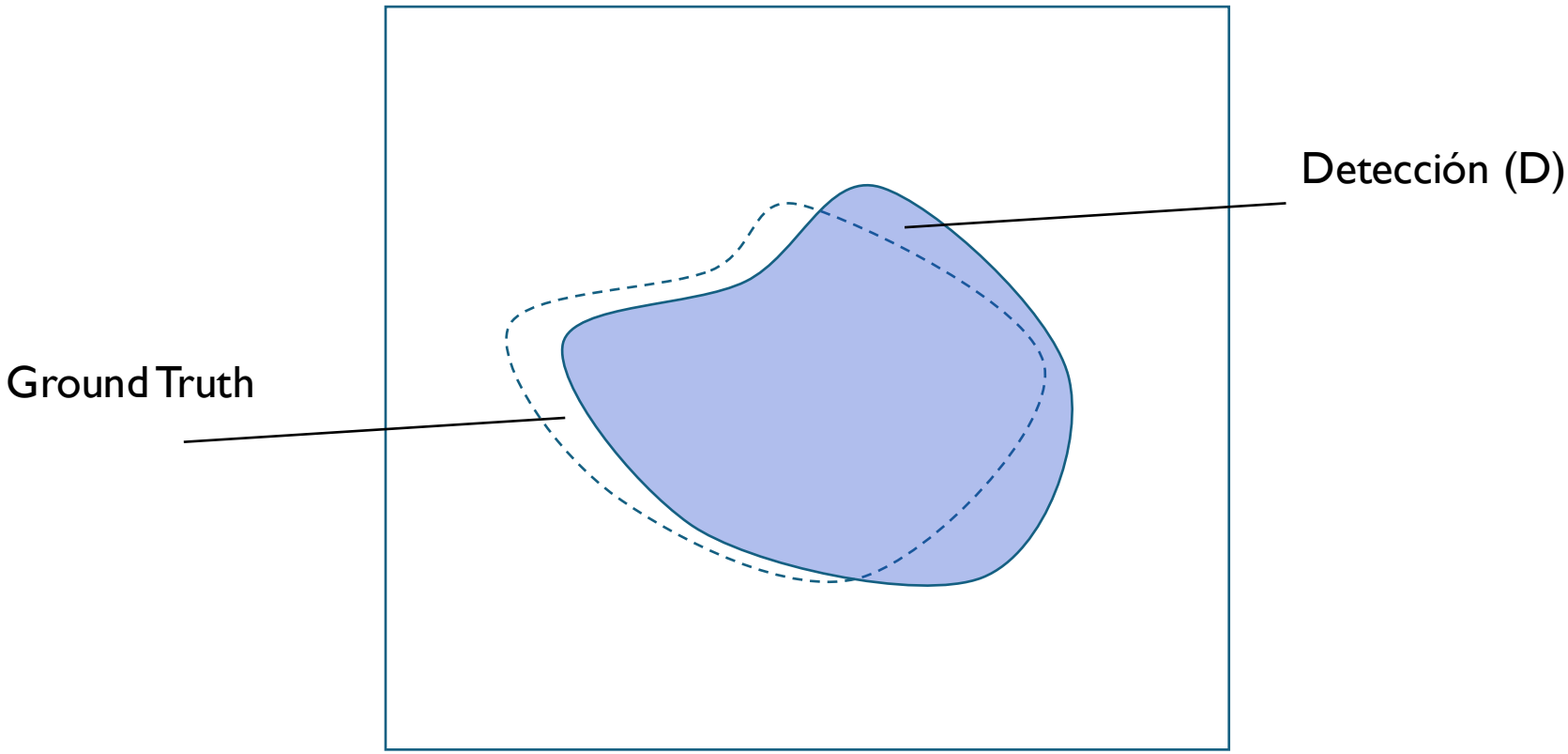


Ejemplo



Detección usando un clasificador

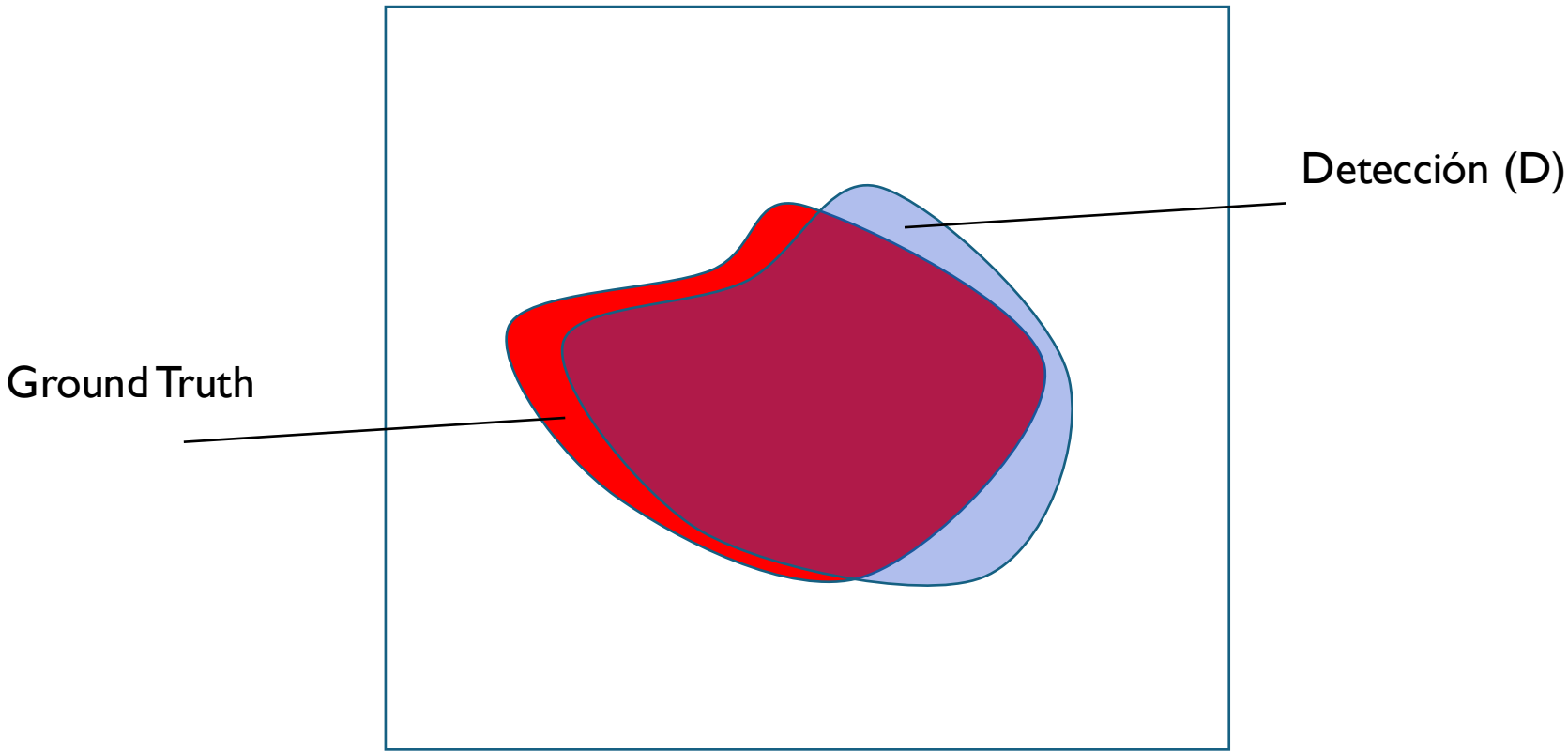
Ejemplo



Detección usando un clasificador

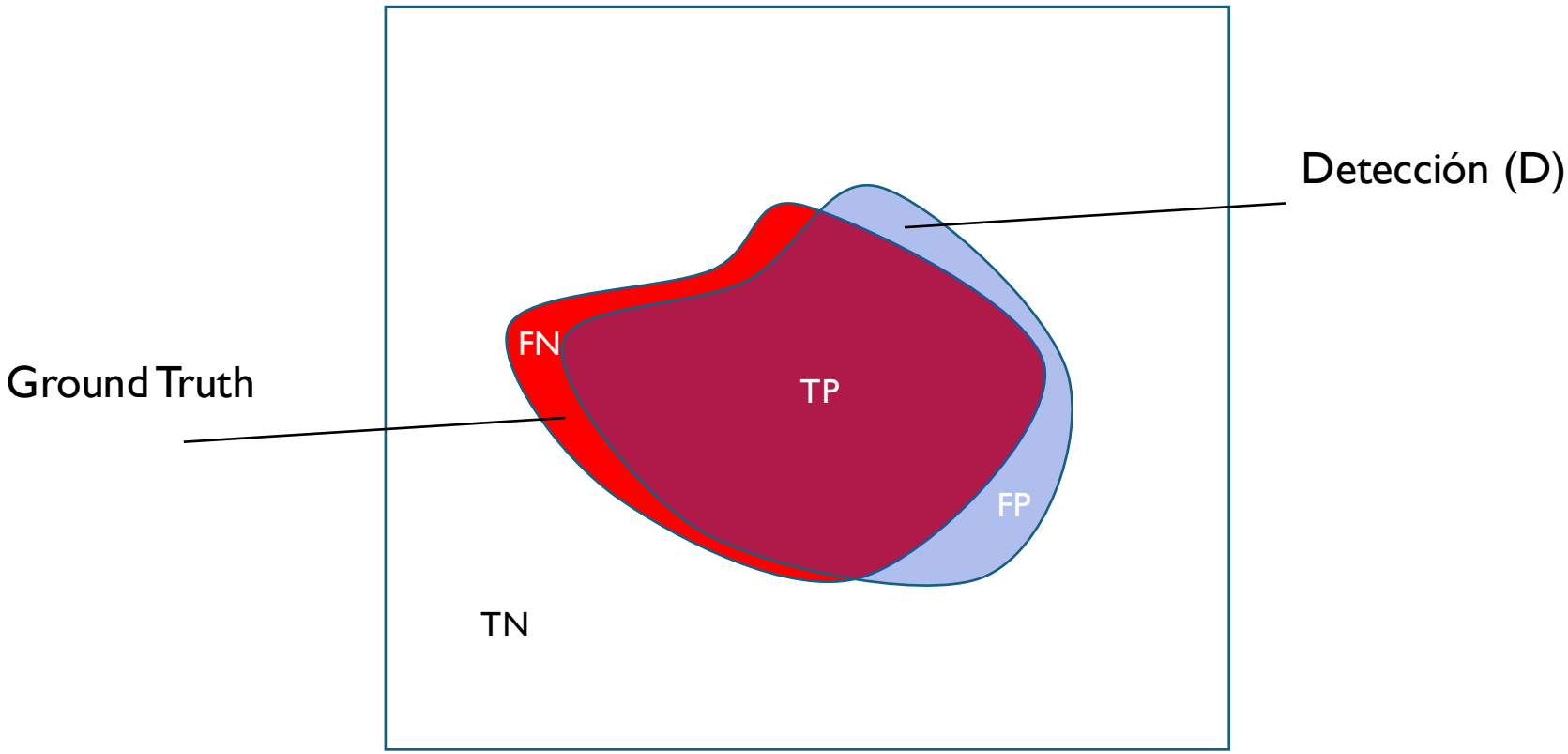
Ejemplo

Píxeles magenta: Verdareos Positivos (TP)
Píxeles azules: falsos positivos (FP)
Píxeles rojos: falsos negativos (FN)
Píxeles blancos: Verdaderos negativos (TN)



Ejemplo

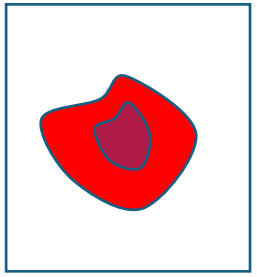
Píxeles magenta: Verdareos Positivos (TP)
Píxeles azules: falsos positivos (FP)
Píxeles rojos: falsos negativos (FN)
Píxeles blancos: Verdaderos negativos (TN)



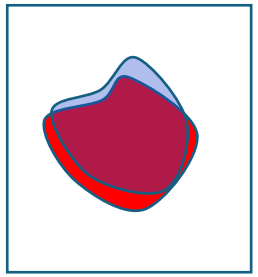
IDEAL
FN = 0
FP = 0

Píxeles magenta: Verdareos Positivos (TP)
Píxeles azules: falsos positivos (FP)
Píxeles rojos: falsos negativos (FN)
Píxeles blancos: Verdaderos negativos (TN)

Extremo
Sin falsos positivos

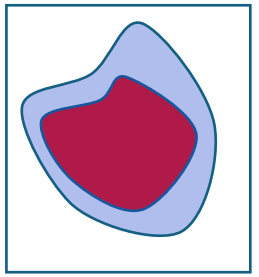


Sin falsas alarmas



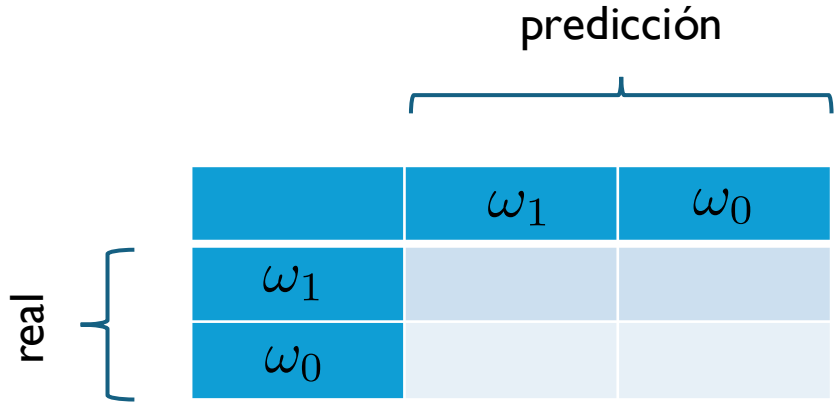
Realidad:
Compromiso entre
FP y FN

Extremo
Se detectan todas las
muestras positivas



Todos los targets
detectados

Matriz de Confusión: dos clases



- True Positive (TP): number of targets correctly classified.
- True Positive (TN): number of non–targets correctly classified.
- False Positive (FP): number of non–targets classified as targets. The false positives are known as 'false alarms' and 'Type I error'.
- False Negative (FN): number of targets classified as no-targets. The false negatives are known as 'Type II error'.

Matriz de Confusión: dos clases

"TARGET DETECTION"

ω_1 Target ω_0 No-Target

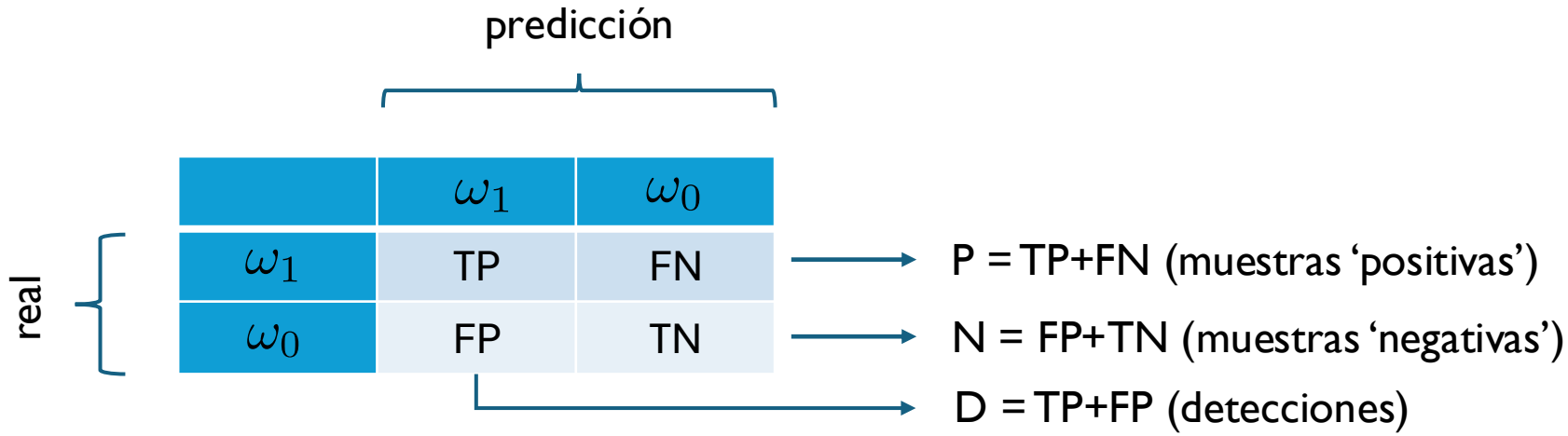
predicción

real

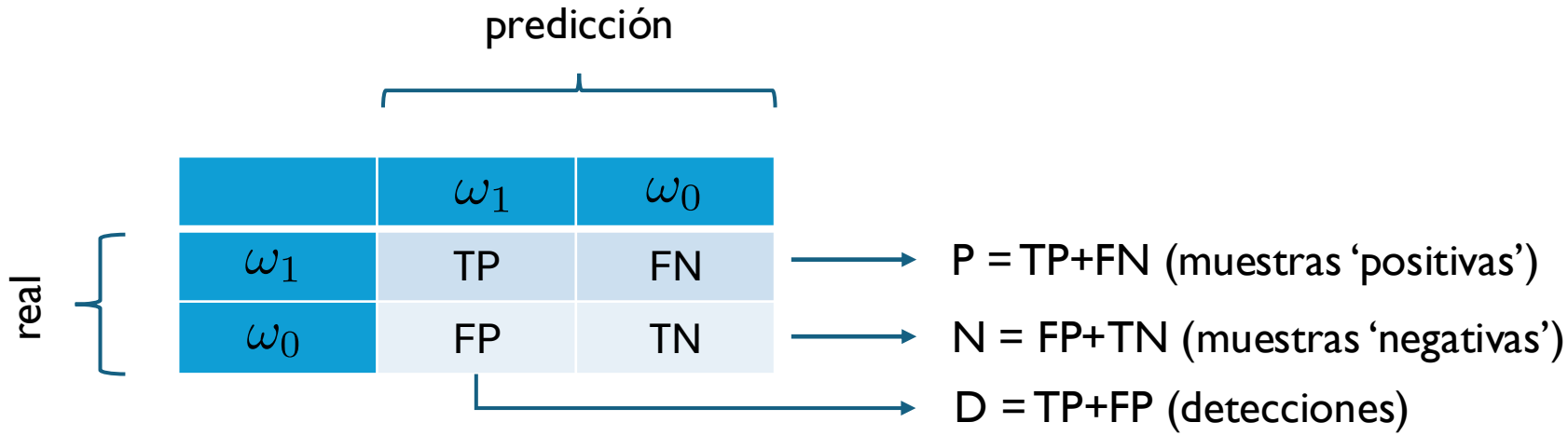
	ω_1	ω_0
ω_1	TP	FN
ω_0	FP	TN

- True Positive (TP): number of targets correctly classified.
- True Positive (TN): number of non–targets correctly classified.
- False Positive (FP): number of non–targets classified as targets. The false positives are known as 'false alarms' and 'Type I error'.
- False Negative (FN): number of targets classified as no-targets. The false negatives are known as 'Type II error'.

Matriz de Confusión: dos clases



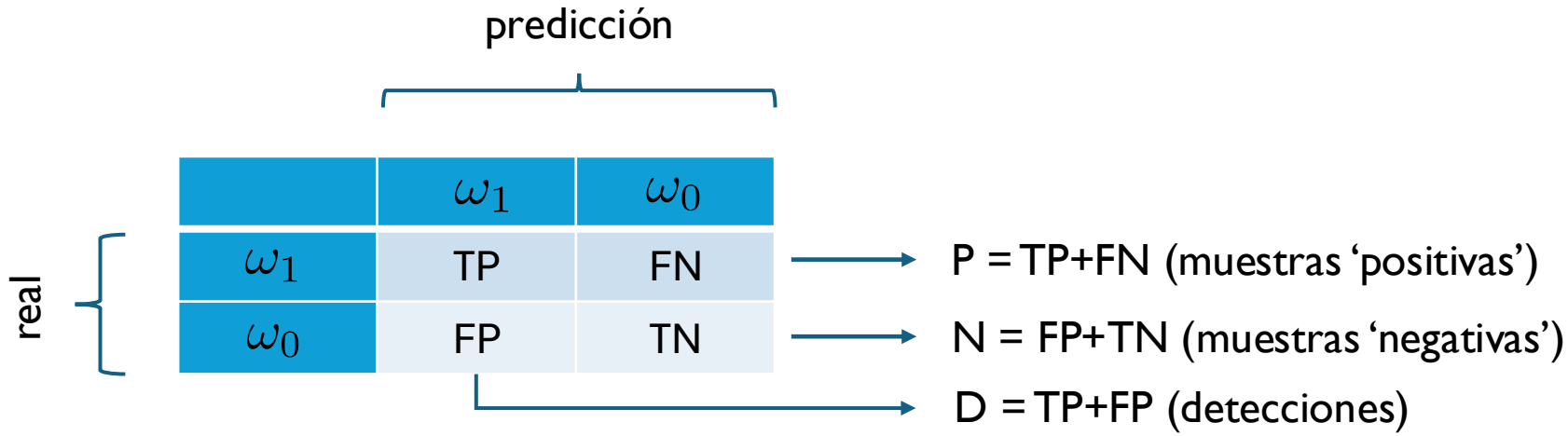
Matriz de Confusión: dos clases



Precision = _____

Recall = _____

Matriz de Confusión: dos clases



$$\text{Precision} = \frac{TP}{TP + }$$

$$\text{Recall} = \frac{TP}{TP + }$$

Matriz de Confusión: dos clases

predicción

real

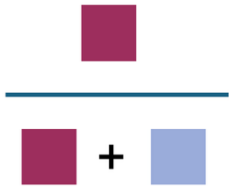
	ω_1	ω_0
ω_1	TP	FN
ω_0	FP	TN

P = TP+FN (muestras 'positivas')

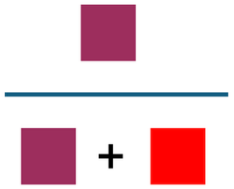
N = FP+TN (muestras 'negativas')

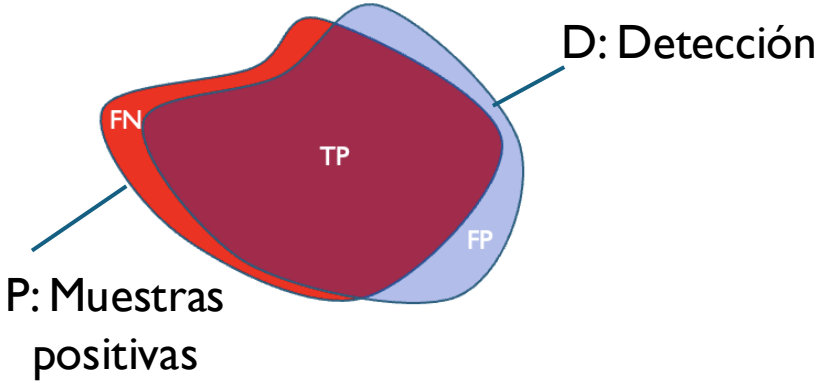
D = TP+FP (detecciones)

$$\text{Precision} = \frac{TP}{TP + FP} =$$



$$\text{Recall} = \frac{TP}{TP + FN} =$$





Definiciones

True positive rate, known as Sensitivity or Recall:

$$TPR = S_n = Re = \frac{TP}{P} = \frac{TP}{TP+FN}$$

Precision or Positive Predictive Value:

$$Pr = \frac{TP}{D} = \frac{TP}{TP+FP}$$

True negative rate, known as Specificity:

$$TNR = Sp = \frac{TN}{N} = \frac{TN}{TN+FP}$$

False positive rate, known as 1-Specificity:

$$FPR = 1 - Sp = \frac{FP}{N} = \frac{FP}{TN+FP}$$

False negative rate, known as Miss Rate:

$$FNR = MR = \frac{FN}{P} = \frac{FN}{TP+FN}$$

Accuracy:

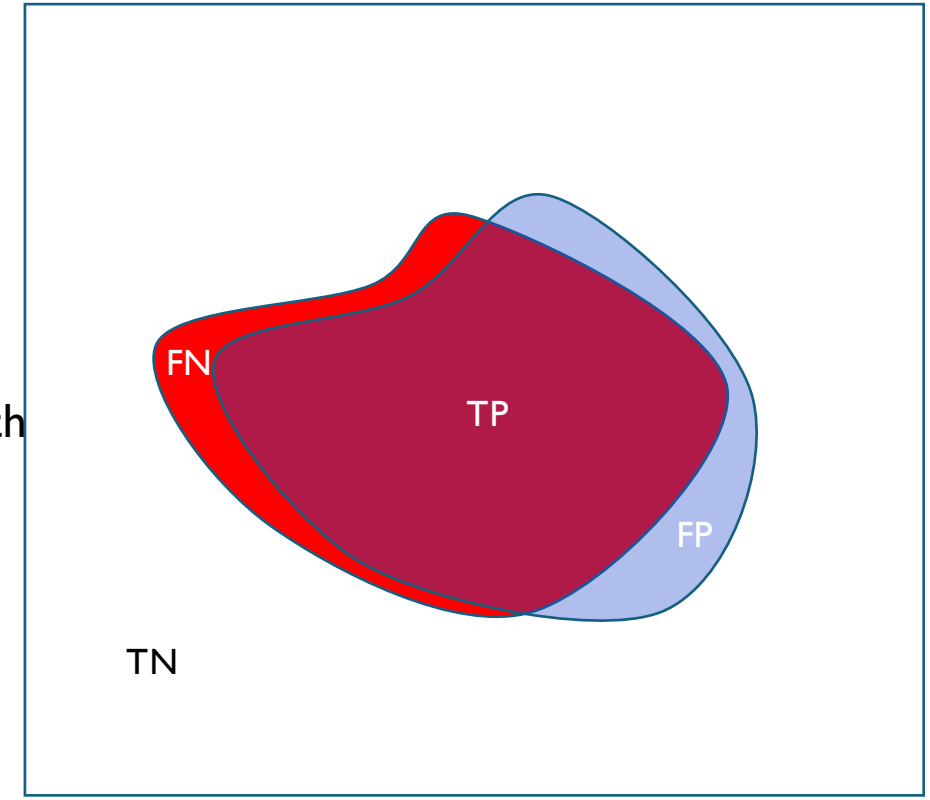
$$ACC = \frac{TP+TN}{P+N}$$

Curva
Precision – Recall

Ejemplo

Píxeles magenta: Verdareos Positivos (TP)
Píxeles azules: falsos positivos (FP)
Píxeles rojos: falsos negativos (FN)
Píxeles blancos: Verdaderos negativos (TN)

Observar que:
TP+FN = Ground Truth
TP+FP = Detección

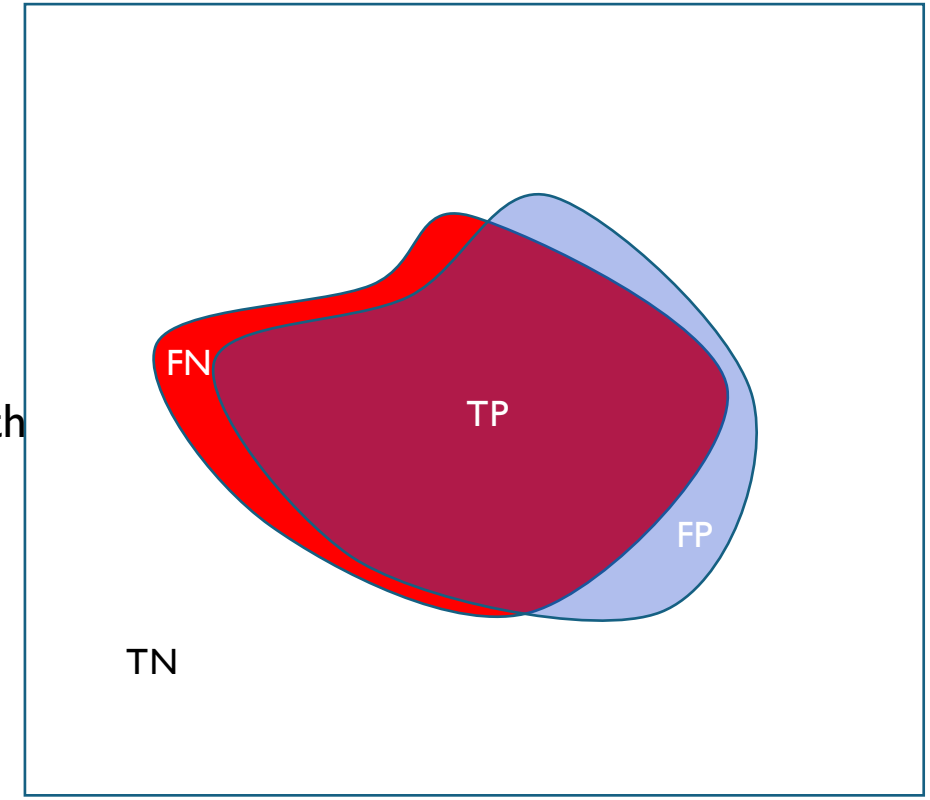


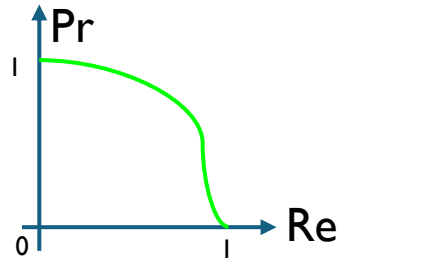
Ejemplo

Píxeles magenta: Verdareos Positivos (TP)
Píxeles azules: falsos positivos (FP)
Píxeles rojos: falsos negativos (FN)
Píxeles blancos: Verdaderos negativos (TN)

Pr = TP/(TP+FP)
Re = TP/(TP+FN)

Observar que:
TP+FN = Ground Truth
TP+FP = Detección

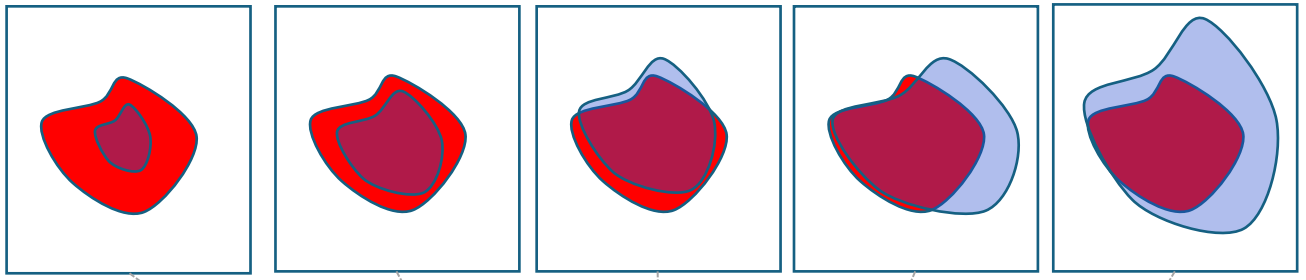


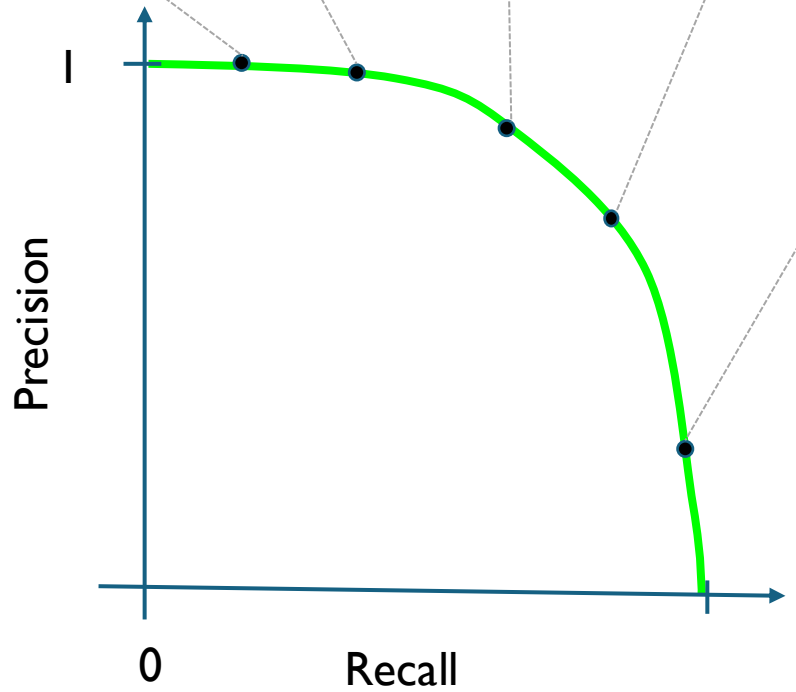


Curva Precision-Recall

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{\blacksquare}{\blacksquare + \blacksquare}$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{\blacksquare}{\blacksquare + \blacksquare}$$





AREA BAJO LA CURVA

mPA

(mean Precision Average)

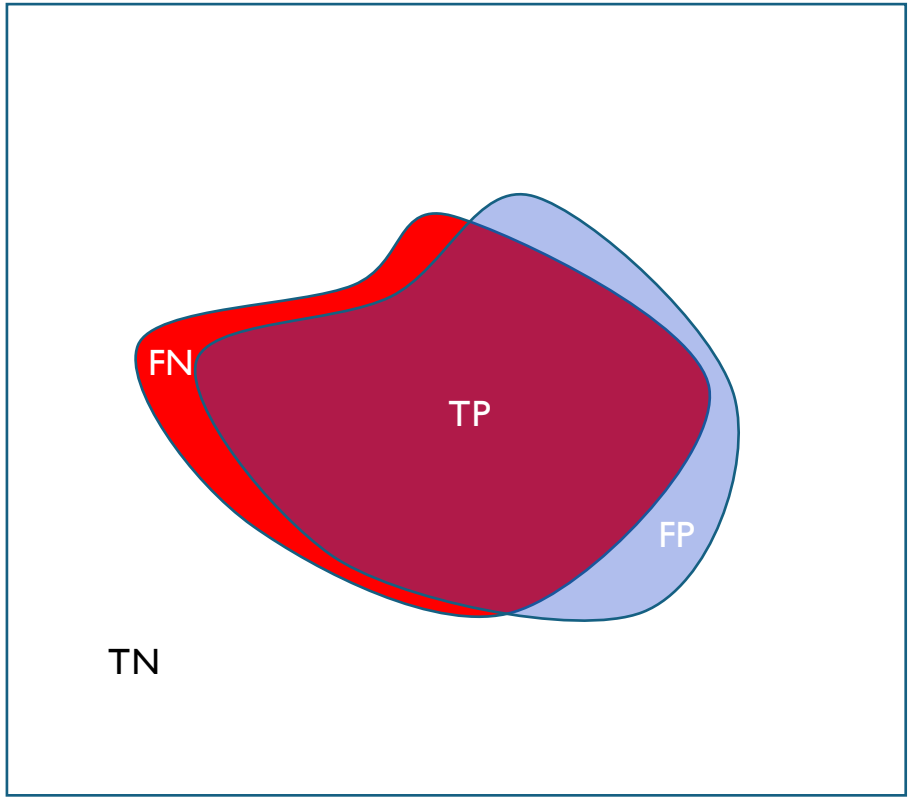
Curva
ROC
(Receiver Operation Characteristic)

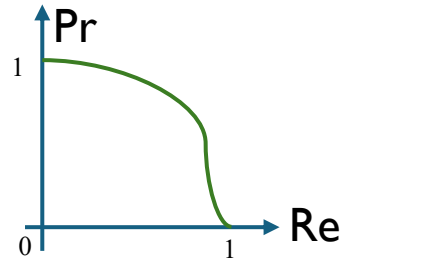
Ejemplo

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Píxeles azules: falsos positivos (FP)
Píxeles rojos: falsos negativos (FN)
Píxeles blancos: Verdaderos negativos (TN)

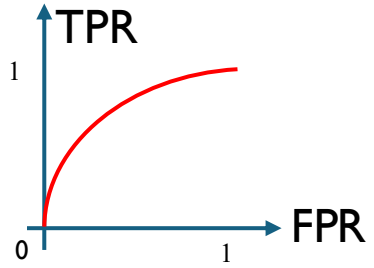
$Pr = TP/(TP+FP)$
$Re = TP/(TP+FN)$

$TPR = TP/(TP+FN)$
$FPR = FP/(TN+FP)$





Curva Precision-Recall



Curva ROC

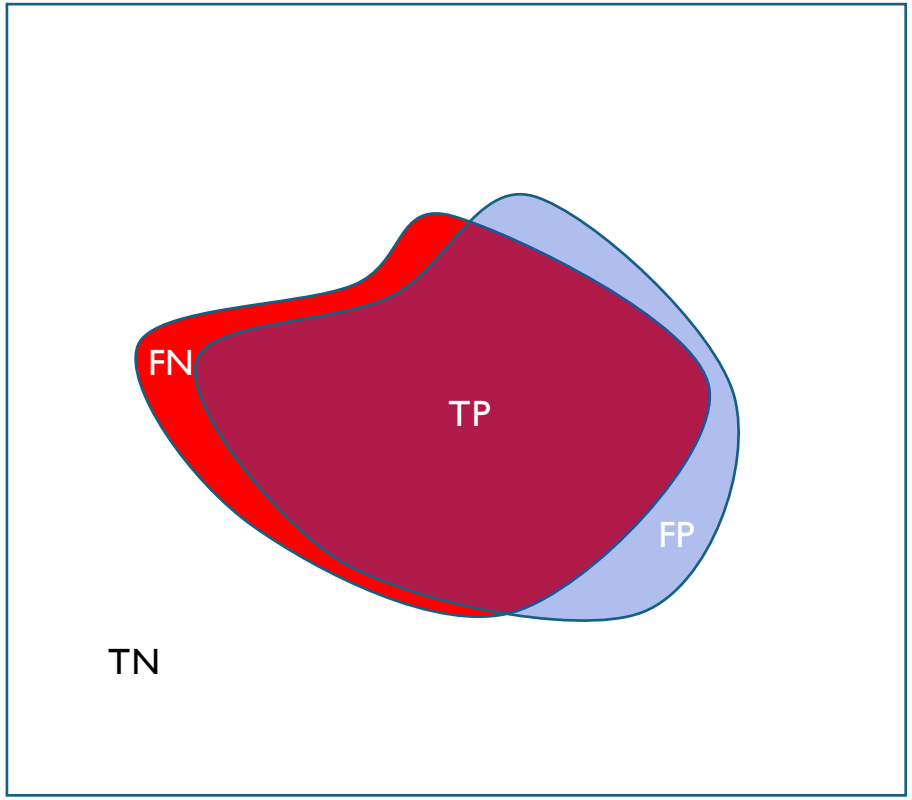
Ejemplo

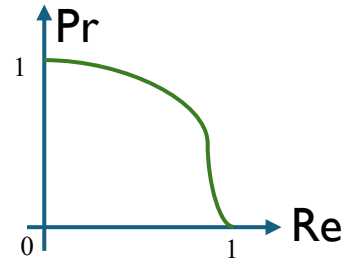
Píxeles magenta: Verdareos Positivos (TP)
Píxeles azules: falsos positivos (FP)
Píxeles rojos: falsos negativos (FN)
Píxeles blancos: Verdaderos negativos (TN)

$Pr = TP/(TP+FP)$
$Re = TP/(TP+FN)$

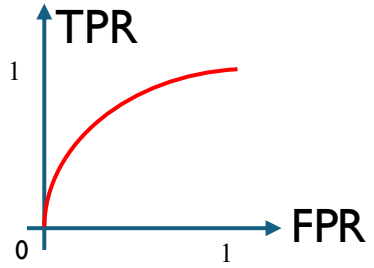
$TPR = TP/(TP+FN)$
$FPR = FP/(TN+FP)$

Sensitivity = TPR
1-Specificity = FPR





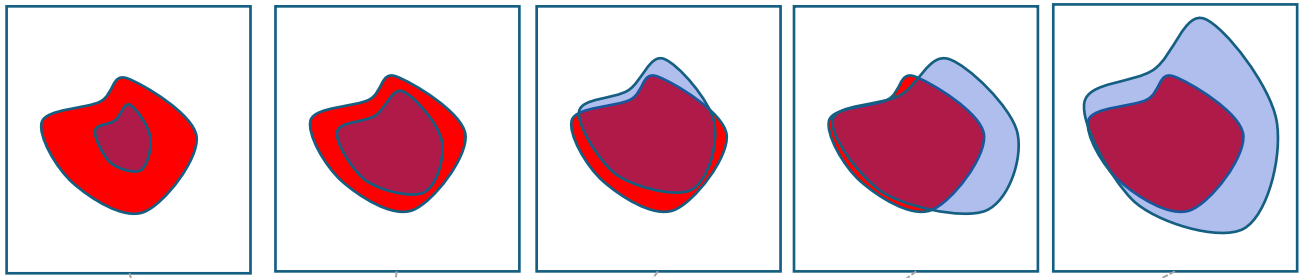
Curva Precision-Recall

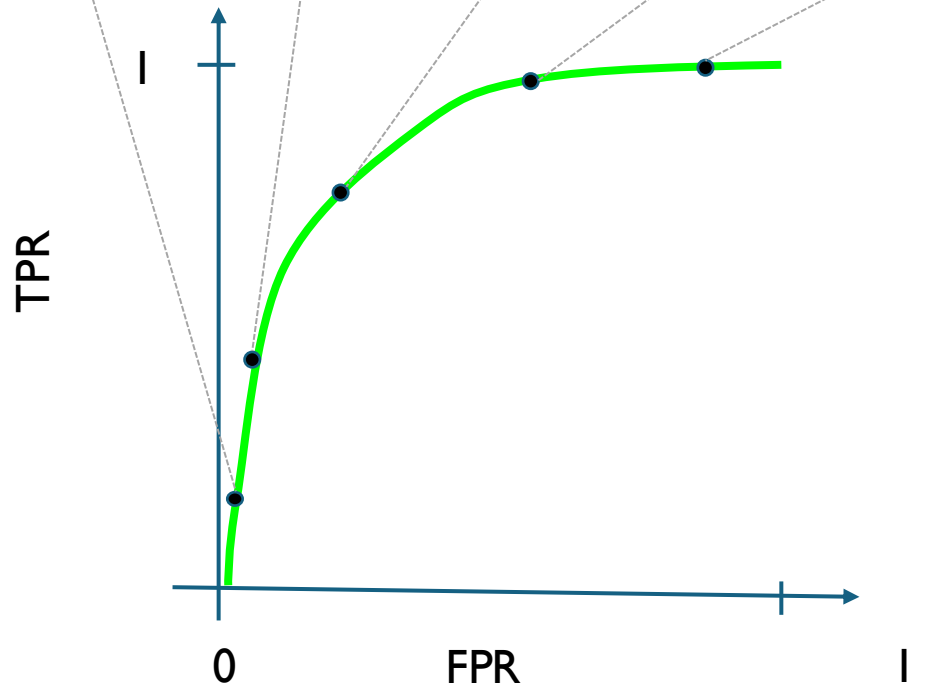


Curva ROC

$$TPR = \frac{TP}{TP+FN} = \frac{\blacksquare}{\blacksquare + \blacksquare}$$

$$FPR = \frac{FP}{TN+FP} = \frac{\blacksquare}{\square + \blacksquare}$$



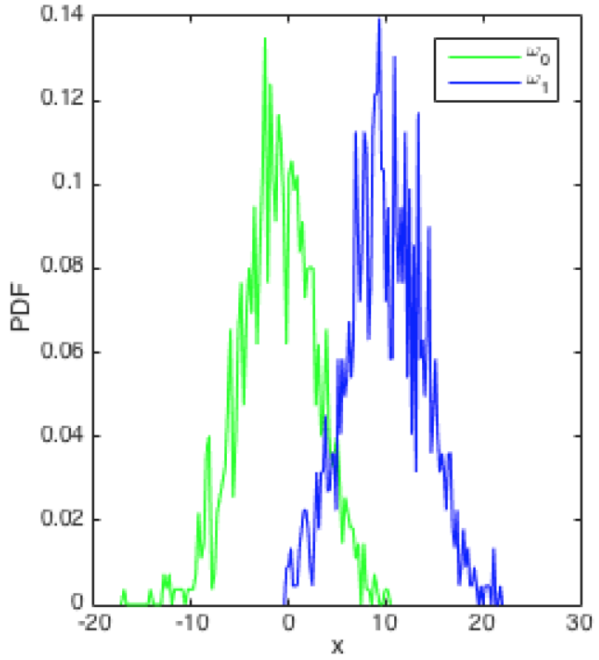


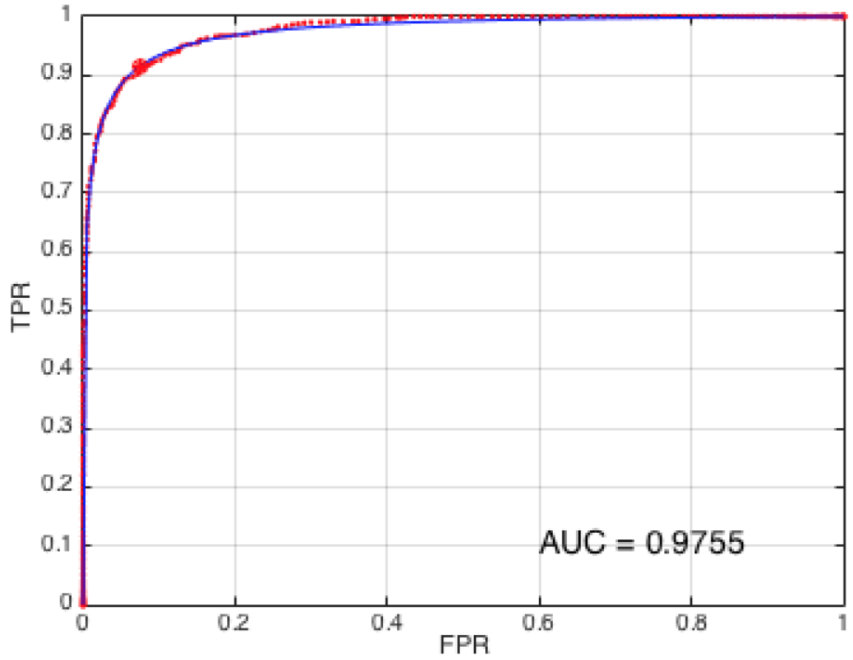
AREA BAJO LA CURVA

Az

(AUC)

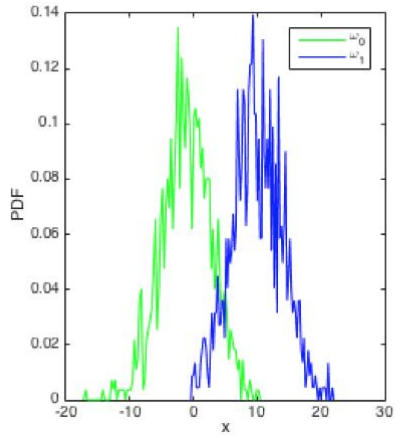
Ejemplo: Curva ROC

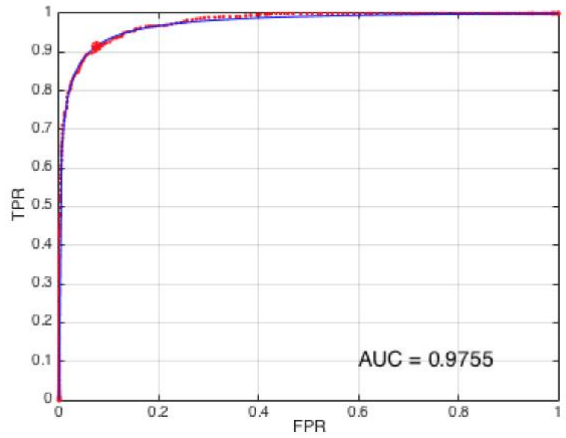


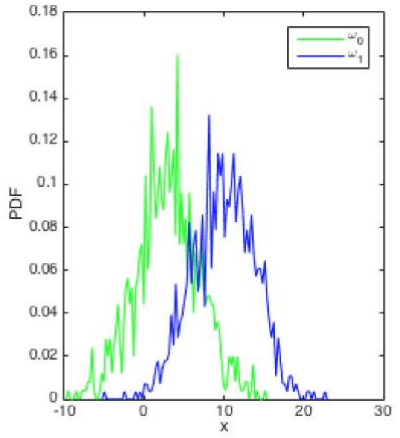


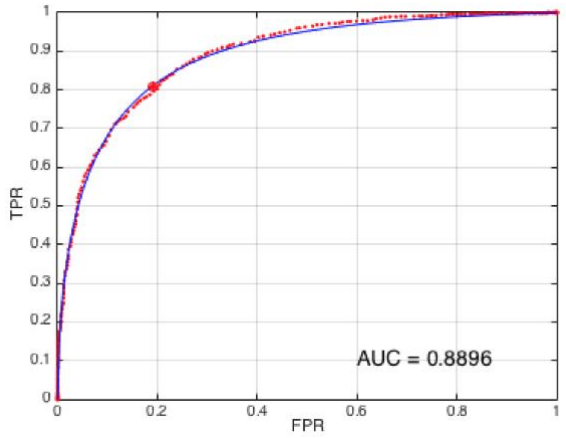
ROC curves (right) for different class distributions (left). The area under the curve (AUC) gives a good measure of the performance of the detection. The obtained points (x_i, y_i) are used to fit the ROC curve to $y = (1 - a^{\gamma x^b})/(1 - a^{\gamma})$. In each ROC curve, the 'best operation point' is shown as *. This point is defined as the closest point to ideal operation point (0,1).

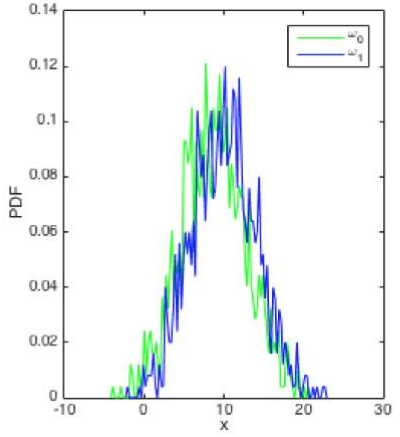
Ejemplo: Curva ROC

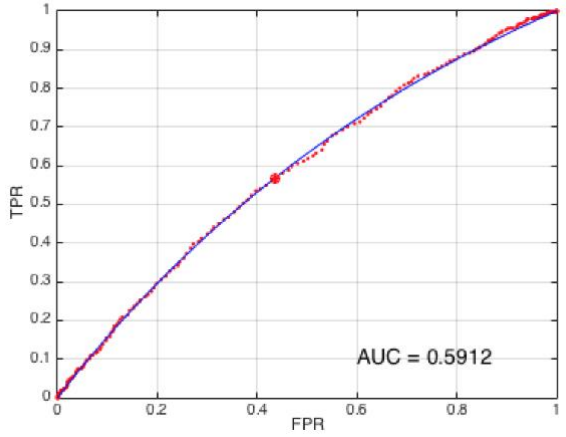












Genuinos & Impostores

(métricas en biometría)

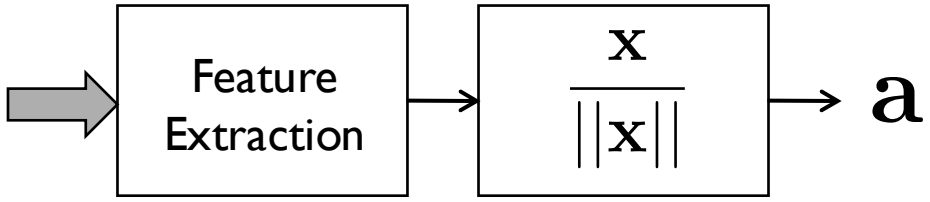
Face Verification Schema

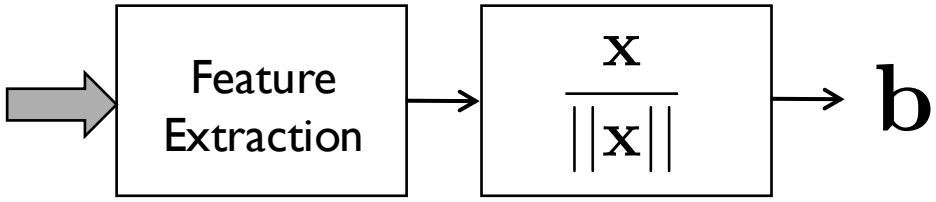


Imagen A

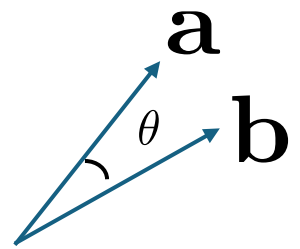


Imagen B





Vectores de n elementos (1 x n)



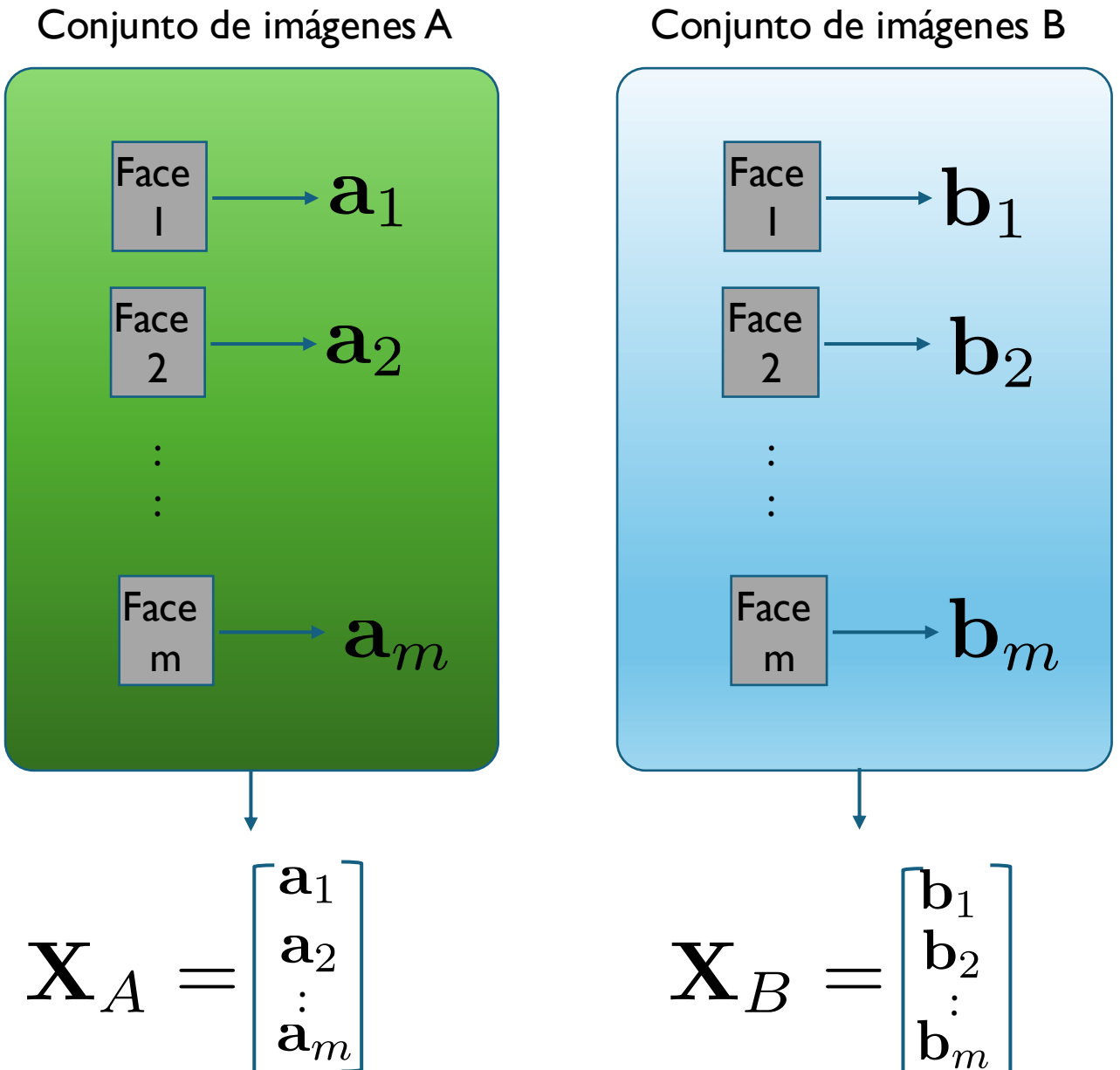
$$s = \cos(\theta)$$

Similitud:
cos - similarity

$$s = \mathbf{a}\mathbf{b}^{\mathsf{T}}$$

Si la imagen-A y la imagen-B son de la misma persona score s es alta

Si la imagen-A y la imagen-B son de personas diferentes score s es baja

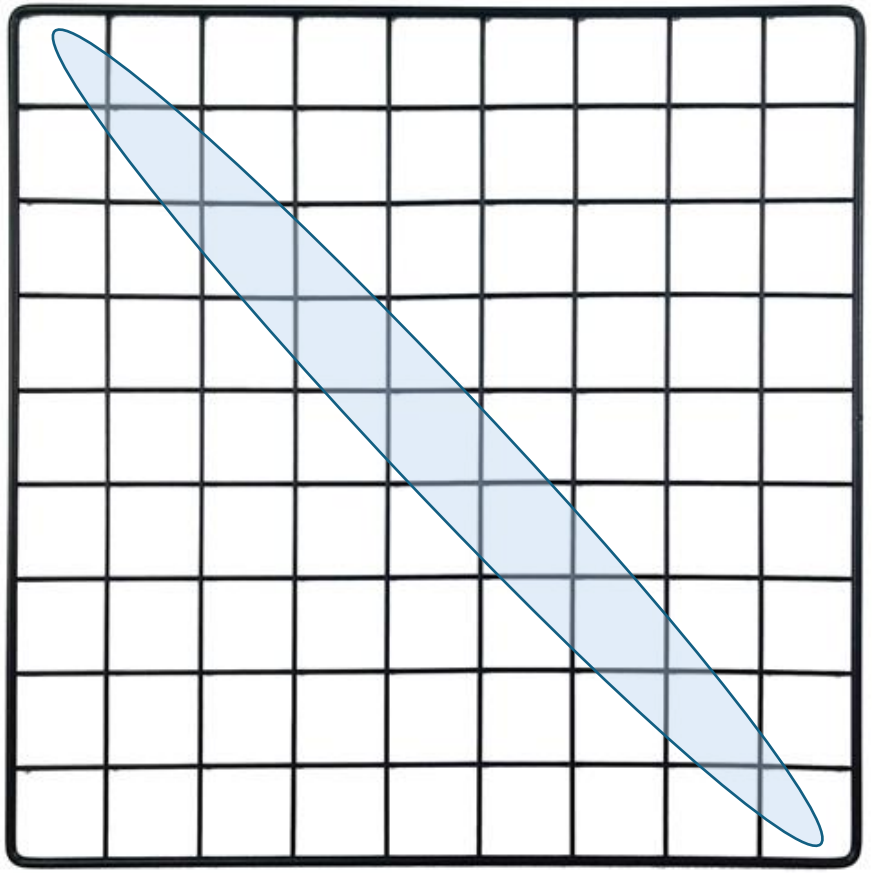


La cara i del conjunto A y la cara i del conjunto B son de la misma persona.

La cara i del conjunto A y la cara j del conjunto B para i=j son de/personas diferentes.

$$\mathbf{S} = \mathbf{X}_A \mathbf{X}_B^{\mathsf{T}} =$$

m x m elementos

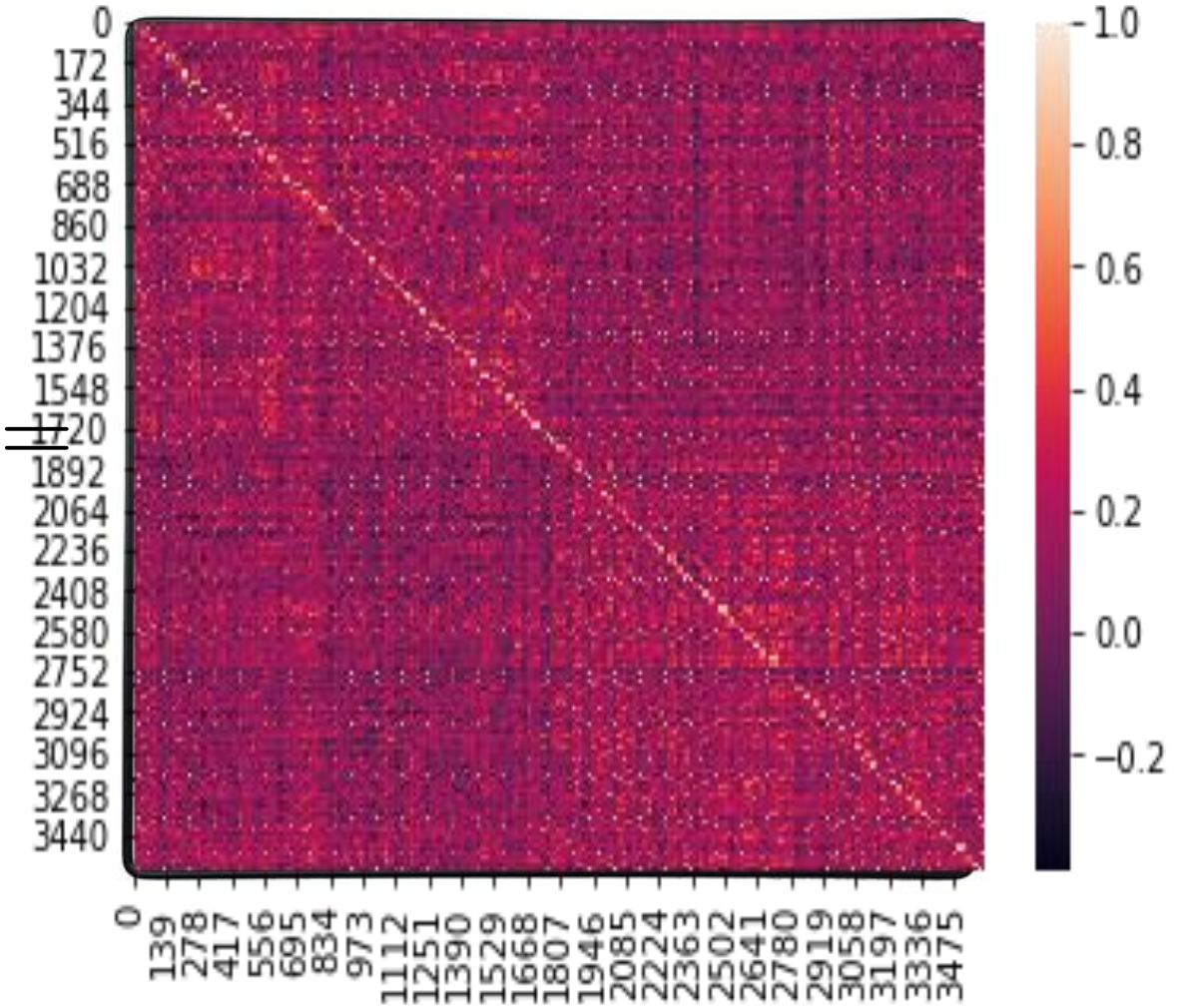


$s_{i,j} =$ Similitud entre la cara i del conjunto A y la cara j del conjunto B

d+: Diagonal: pares positivos (genuines)
d-: No Diagonal: pares negativos (impostores)

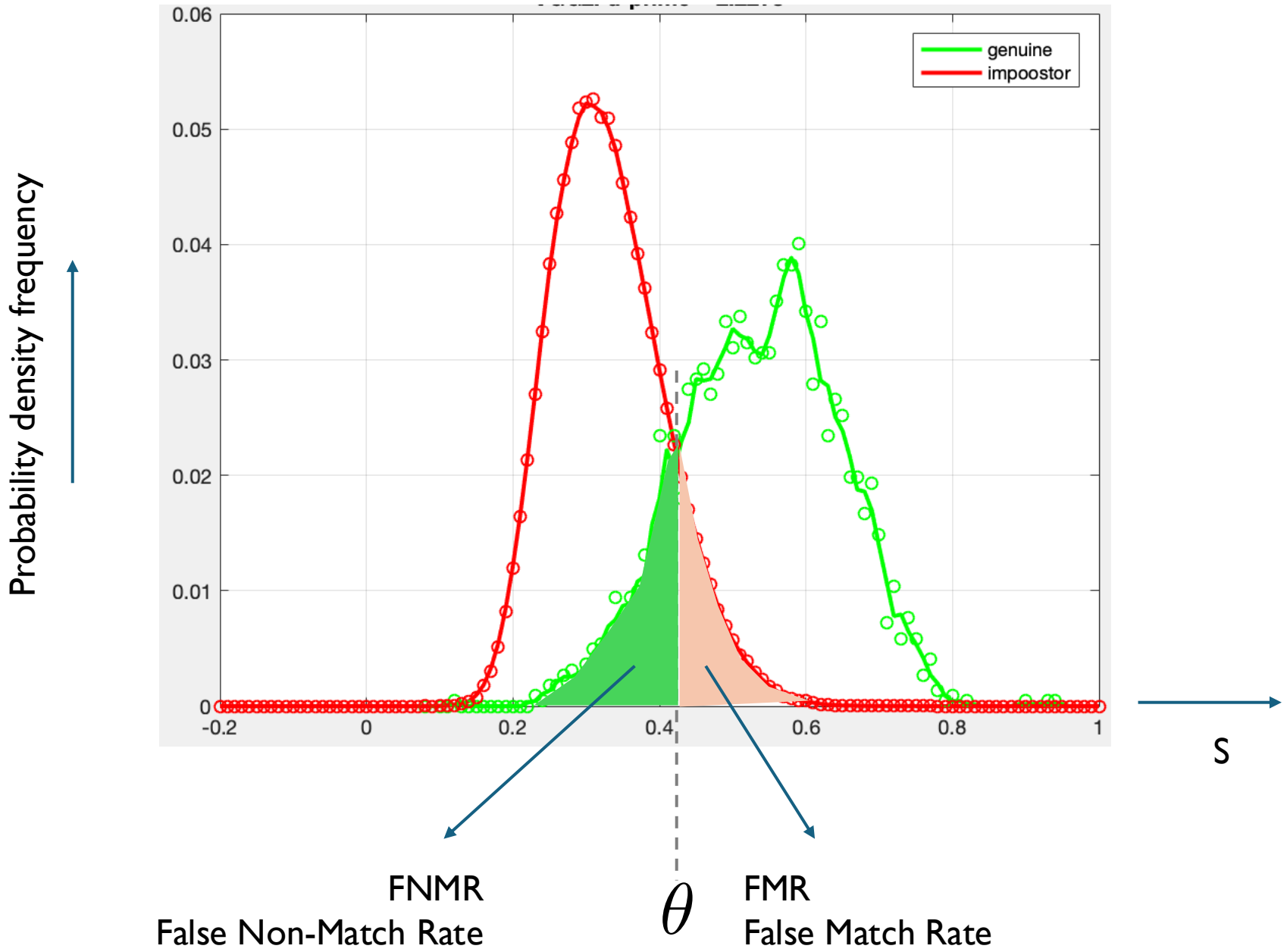
$$\mathbf{S} = \mathbf{X}_A \mathbf{X}_B^\top =$$

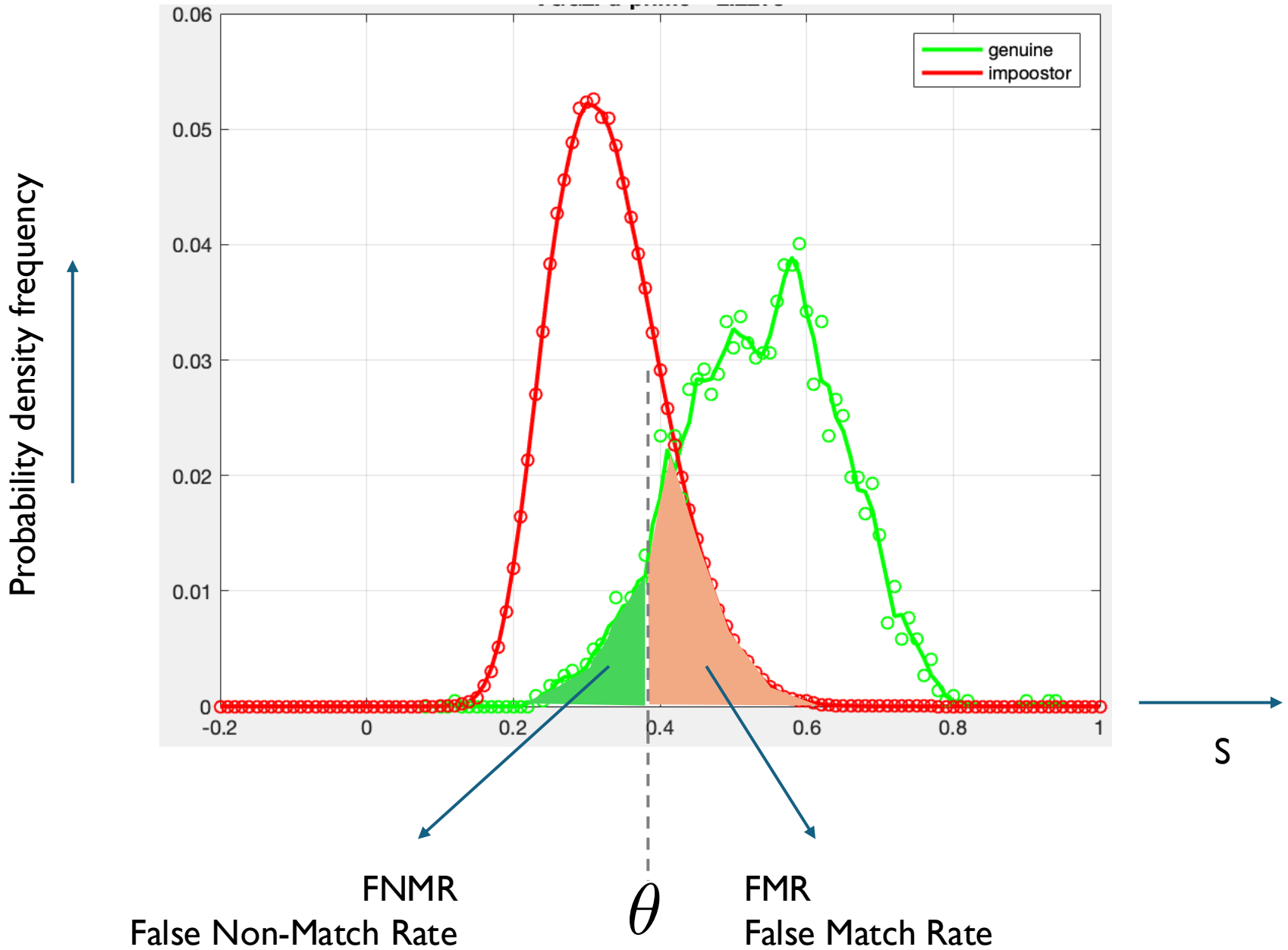
m x m elementos

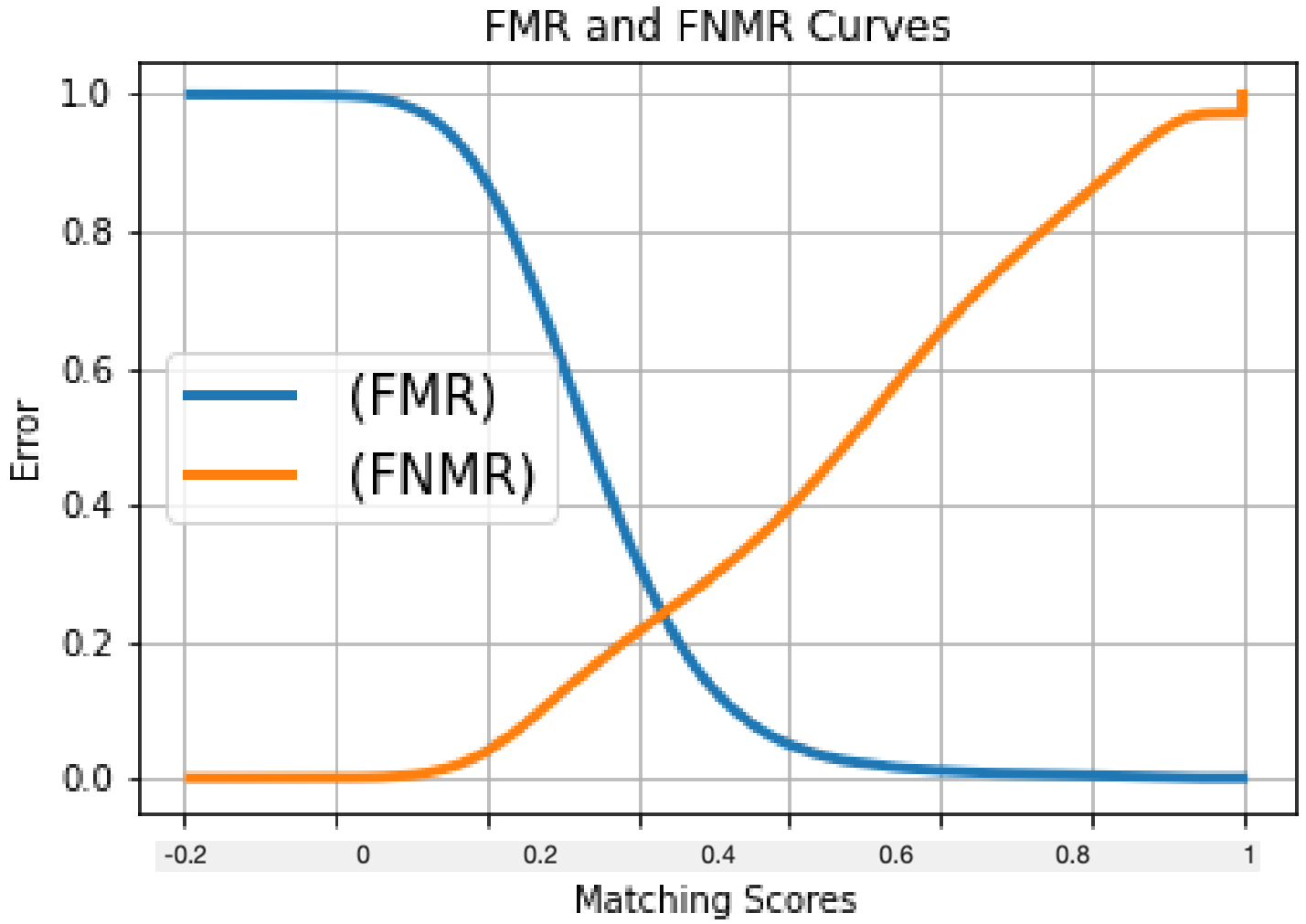


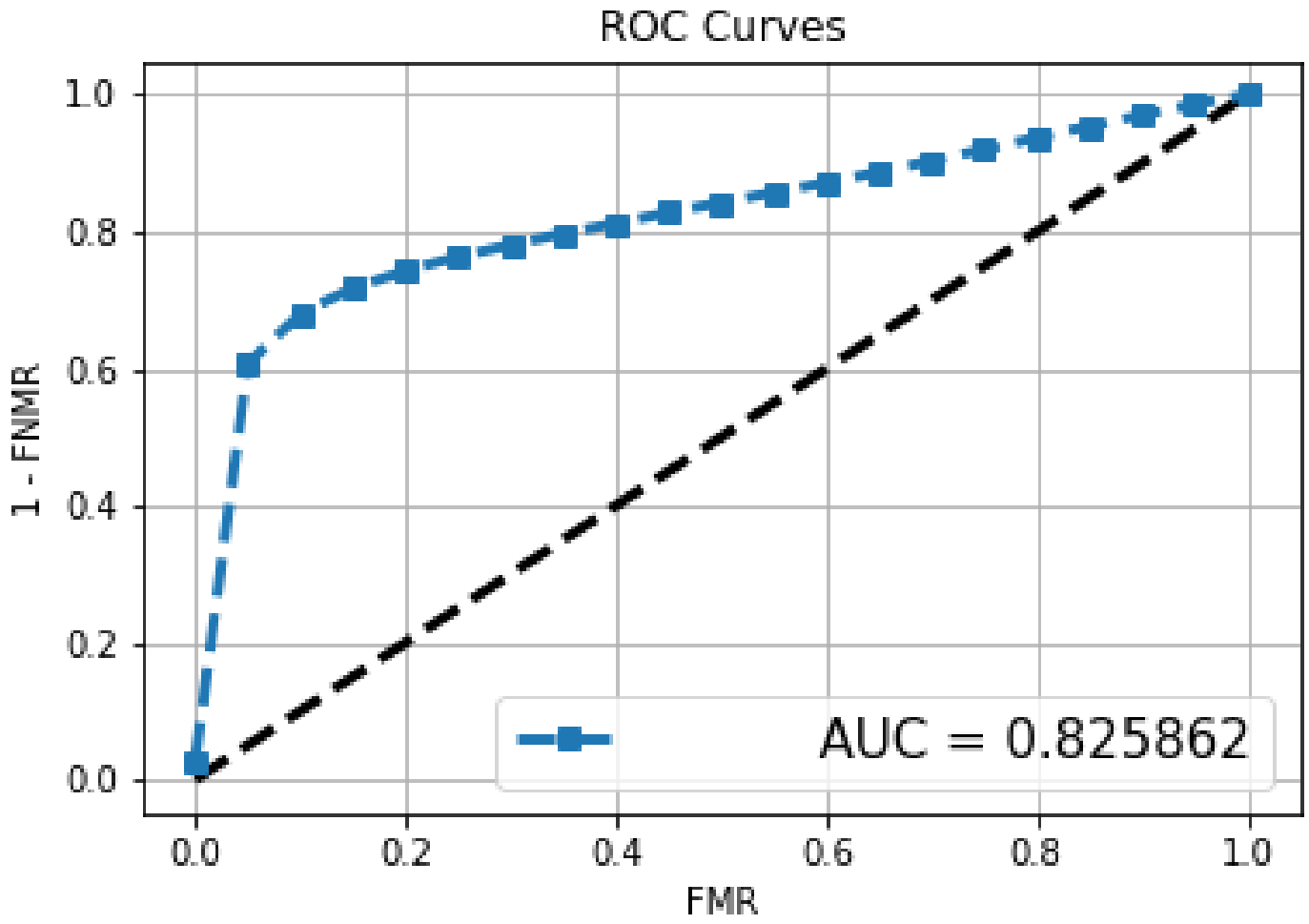
$s_{i,j} =$ Similitud entre la cara i del conjunto A y la cara j del conjunto B

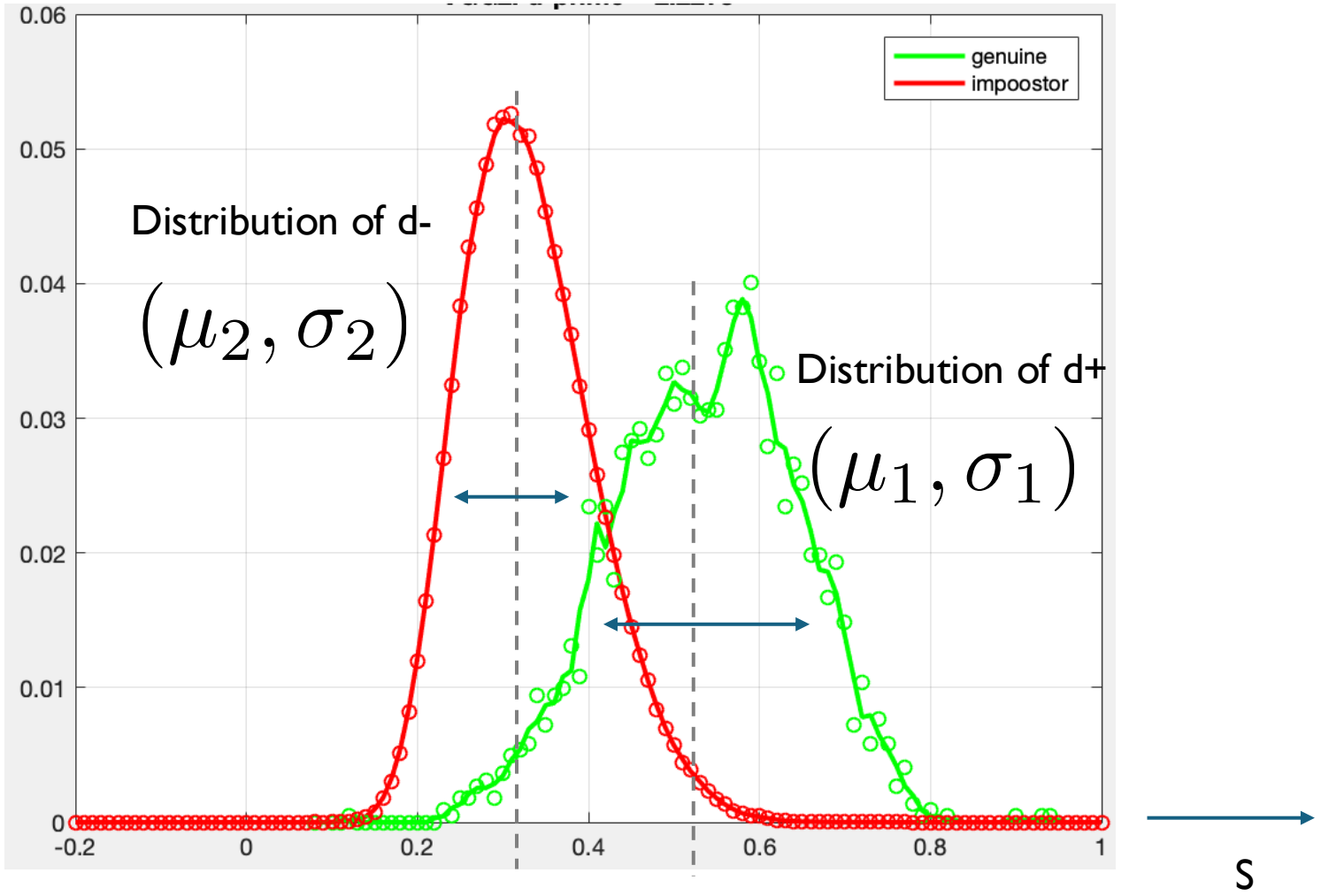
d+: Diagonal: pares positivos (genuines)
d-: No Diagonal: pares negativos (impostores)


Probability density frequency
0.06
0.05
0.04
0.03
0.02
0.01
0
-0.2
0
0.2
0.4
0.6
0.8
1
genuine
impoostor
S
FNMR
False Non-Match Rate
θ
FMR
False Match Rate


Probability density frequency
0.06
0.05
0.04
0.03
0.02
0.01
0
-0.2
0
0.2
0.4
0.6
0.8
1
genuine
impoostor
S
FNMR
False Non-Match Rate
θ
FMR
False Match Rate


FMR and FNMR Curves
1.0
0.8
0.6
0.4
0.2
0.0
Error
(FMR)
(FNMR)
-0.2
0
0.2
0.4
0.6
0.8
1
Matching Scores


ROC Curves
1.0
0.8
0.6
0.4
0.2
0.0
1 - FNMR
0.0
0.2
0.4
0.6
0.8
1.0
FMR
AUC = 0.825862

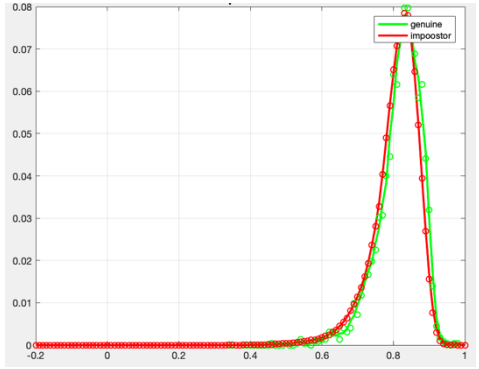


Performance:

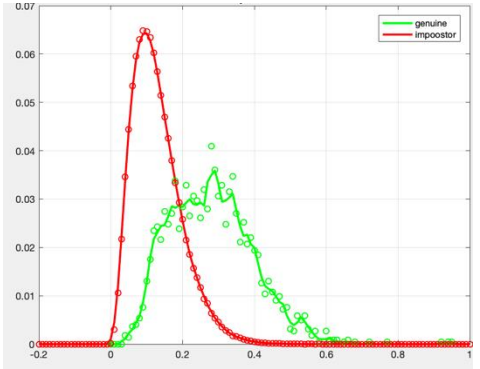
$$d' = \frac{|\mu_1 - \mu_2|}{\sqrt{\frac{1}{2}(\sigma_1^2 + \sigma_2^2)}}$$

Área de cada distribución = 1

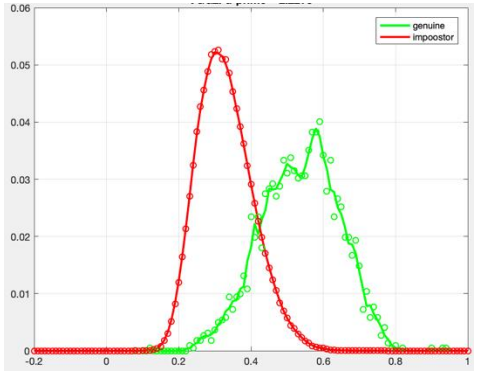
Ejemplos de d'
para diferentes algoritmos



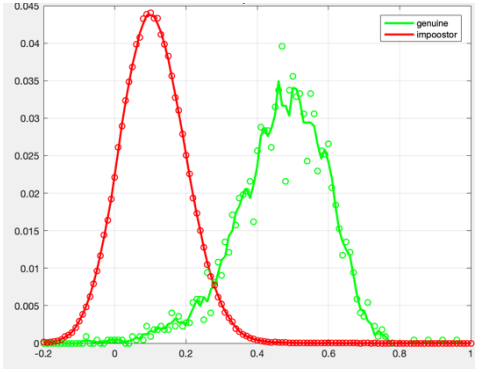
d' = 0.20



d' = 1.59



d' = 2.22



d' = 3.23

The best one